SIMATS ENGINEERING
ASSESSMENT TOOL

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13**

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:25**
Duration : 1 Hour

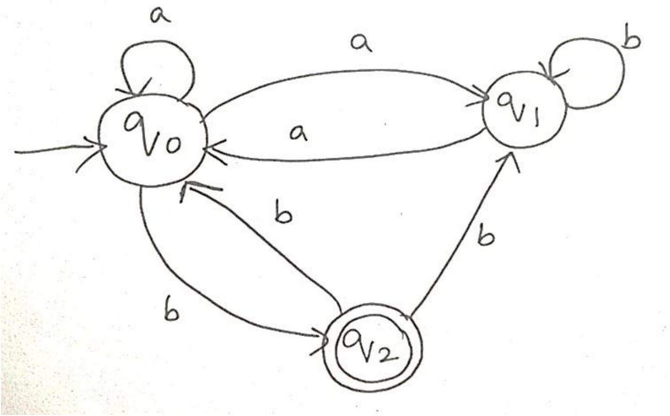
Analytical Problem

Code of Conduct:

I U.Divya/192372277(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Divya*

1. Consider the NFA given below which accepts some language L over the alphabet ={a,b}. Construct a DFA that represents the same language L



From the given NFA:

- **States:** Q = {q0, q1, q2}
- **Alphabet:** Σ = {a, b}
- **Start State:** q0
- **Final State:** q2

Step 1: NFA Transition Table

State	a	b
→q0	{q0,q1}	{q2}
q1	{q0}	{q1}
*q2	∅	{q0,q1}

Step 2: Subset Construction

The DFA states are subsets of the NFA states.

DFA State	NFA State
A	{q0}
B	{q0,q1}
C	{q2}
D	{q0,q1,q2}
E	∅ (Dead State)

Accepting DFA states are those containing **q2**:

Final States = {C, D}

Step 3: DFA Transition Table

DFA State	a	b

->A={q0}	B	C
B={q0,q1}	B	D
*C={q2}	E	B
*D={q0,q1,q2}	B	D
E=∅	E	E

Formal Definition of the DFA
M=(Q,Σ,δ,q0,F)M=(Q,\Sigma,\delta,q_0,F)M=(Q,Σ,δ,q0 ,F)

where

- **Q = {A, B, C, D, E}**
- **Σ = {a, b}**
- **Start State = A**
- **Final States = {C, D}**

Final Answer

DFA Transition Table

DFA State	a	b
->A={q0}	B	C
B={q0,q1}	B	D
*C={q2}	E	B
*D={q0,q1,q2}	B	D
E=∅	E	E

DFA States

- A = {q0}
- B = {q0,q1}
- C = {q2}
- D = {q0,q1,q2}
- E = ∅ (Dead State)

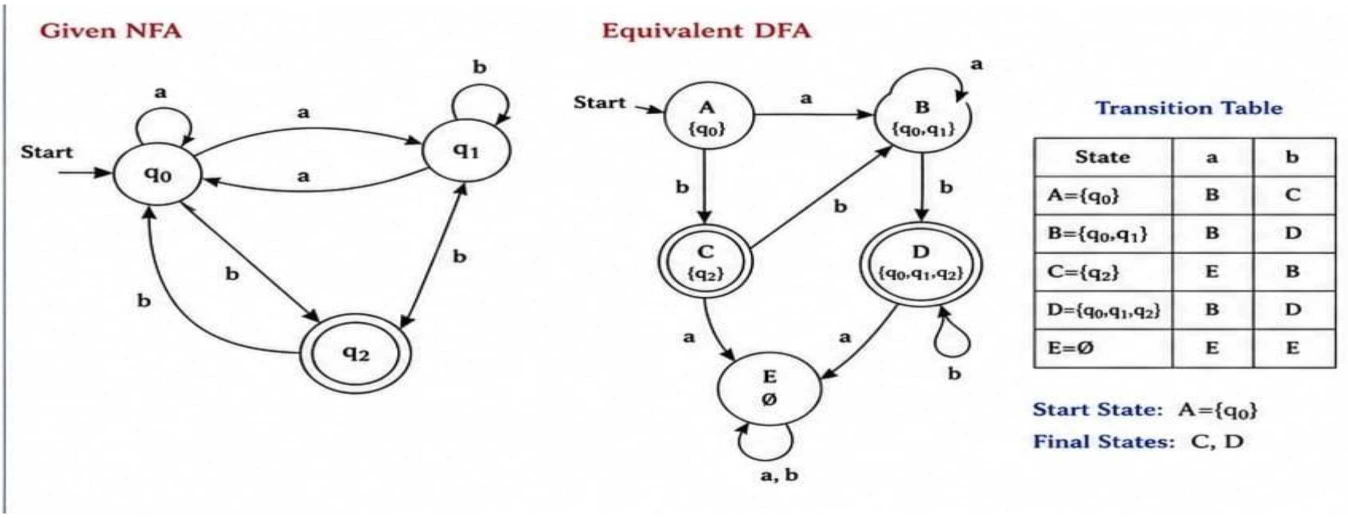
Start State

A

Accepting States

C, D

This solution follows the **subset construction algorithm** for converting the given NFA into an equivalent DFA



2. Let L be the language of words of length atleast 2 whose last two letters are difference from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA

Alphabet

Σ={a,b}\Sigma=\{a,b\}Σ={a,b}

Language

Strings of length **at least 2** ending in **ab** or **ba**.

Examples:

- Accepted: **ab, ba, aab, bba, abaab**
- Rejected: **aa, bb, a, b**

NFA

States

- q0 – Start State
- q1 – Last symbol a
- q2 – Last symbol b
- q3 – Accept State

Transition Table

State	a	b
->q0	{q1}	{q2}
q1	{q1}	{q3}
q2	{q3}	{q2}
*q3	{}	{}

Equivalent DFA

States

A = {q0}

B = {q1}

C = {q2}

D = {q1,q3} (Accept)

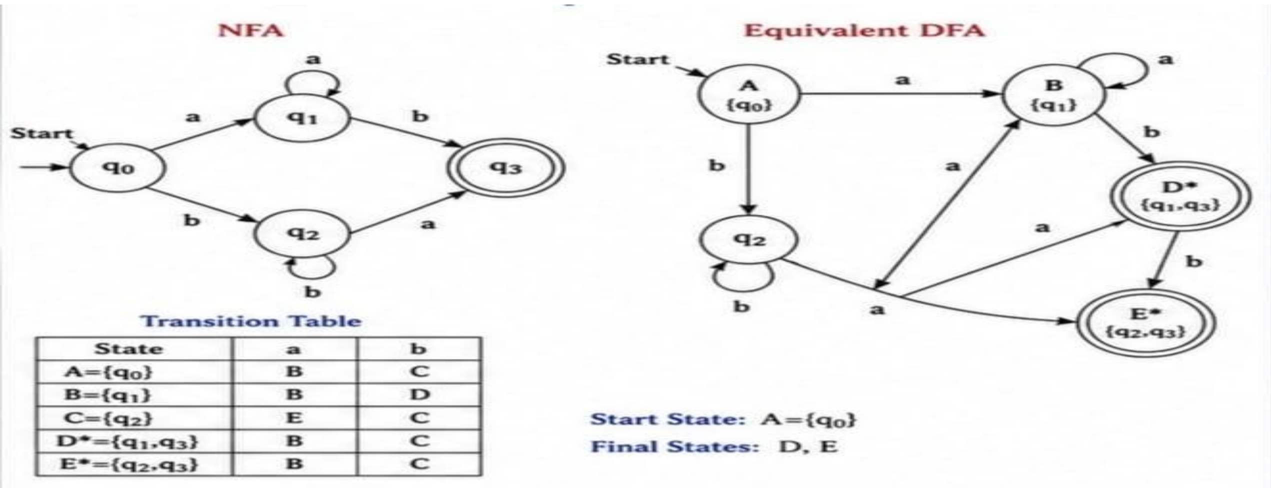
E = {q2,q3} (Accept)

DFA Transition Table

State	a	b
->A	B	C
B	B	D
C	E	C
*D	B	C
*E	B	C

Final States

D, E



3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

Alphabet

Σ={0,1}\Sigma=\{0,1\}Σ={0,1}

Formal Definition

M=(Q,Σ,δ,q0,F)M=(Q,\Sigma,\delta,q_0,F)M=(Q,Σ,δ,q0 ,F)

where

Q={q0,q1,q2,q3,q4}

Σ={0,1}

Start State=q0

Final States={q2,q4}

Transition Function

State	0	1
->q0	{q0,q1}	{q0,q3}
q1	{q2}	{}
*q2	{}	{}
q3	{}	{q4}
*q4	{}	{}

Verification for String 01011

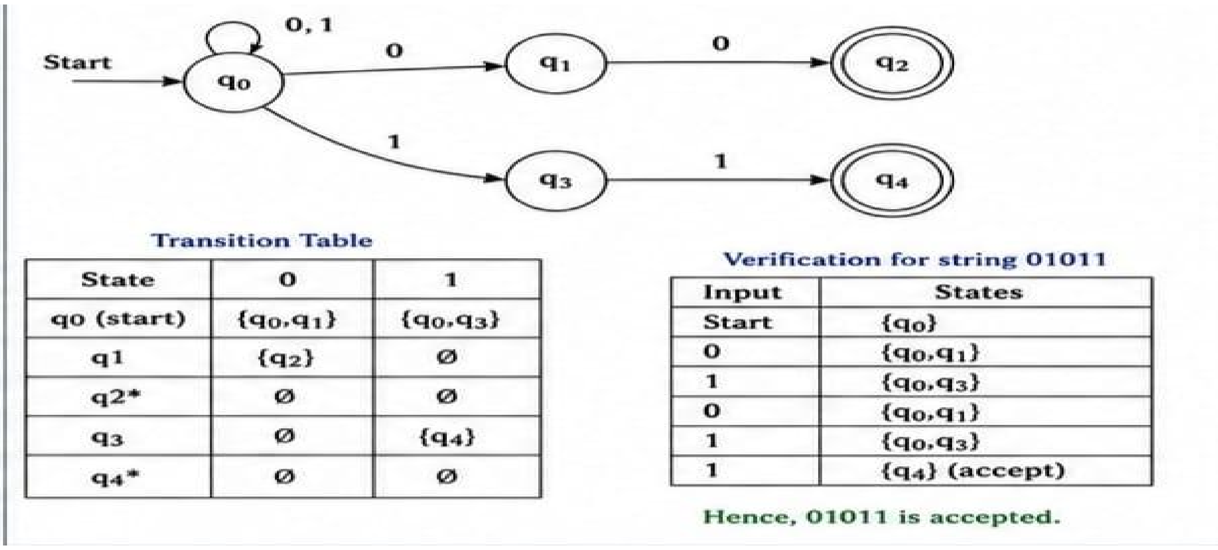
Input	Current State
Start	q0
0	q0,q1
1	q0,q3
0	q0,q1
1	q0,q3
1	q4

Final state reached:

q4

Hence,

01011 is Accepted.



4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet ={0,1}. Construct a DFA that represents the same language.

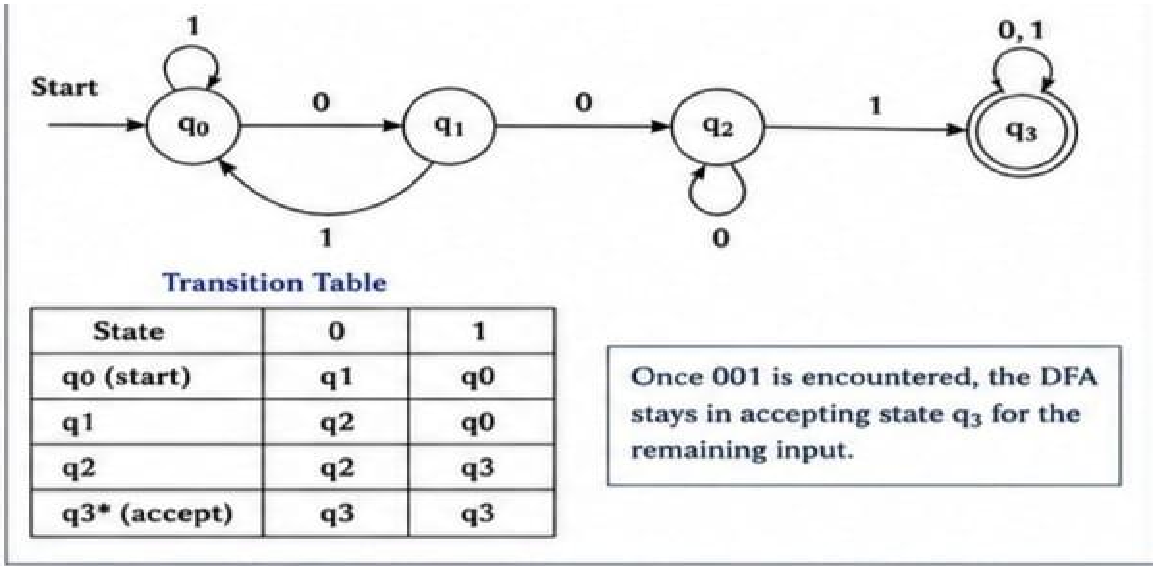
States

- q0 – Start
- q1 – Found first 0
- q2 – Found 00
- q3 – Found 001 (Accept)

Transition Table

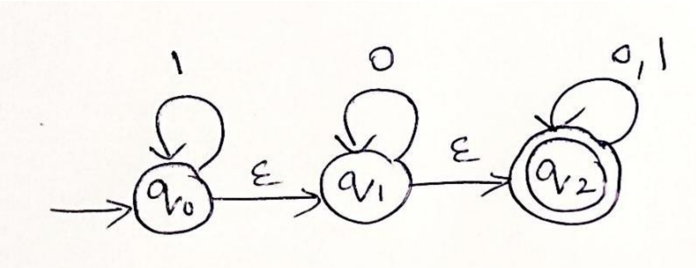
State	0	1
->q0	q1	q0
q1	q2	q0
q2	q2	q3
*q3	*q3	q3

Once the automaton reaches **q3**, every remaining input keeps it in the accepting state because the substring **001** has already been found.



State	0	1
q0 (start)	q1	q0
q1	q2	q0
q2	q2	q3
q3* (accept)	q3	q3

5. A NFA with ε-transitions is given below. Find ε-closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



Step 1: Given ε-NFA

States

Q = {q0, q1, q2}

Alphabet

Σ = {0,1}

Start State

q0

Final State

F = {q2}

Step 2: ε-Closure of Each State

ε-Closure(q0)

From q0 we can move

q0 →ε q1

q1 →ε q2

Therefore,

ε-Closure(q0) = {q0, q1, q2}

ε-Closure(q1)

From q1

q1 →ε q2

Therefore,

ε-Closure(q1) = {q1, q2}

ε-Closure(q2)

No ε-transition.

Therefore,

ε-Closure(q2) = {q2}

Step 3: DFA States

DFA State	NFA State
A	{q0,q1,q2}
B	{q1,q2}
C	{q2}
D	∅ (Dead State)

Since every set containing **q2** is accepting,

Final States = {A, B, C}

Step 4: DFA Transition Table

DFA State	0	1

->*A	B	A
*B	B	C
*C	C	C
D=∅	D	D

Formal Definition of the DFA

M=(Q,Σ,δ,A,F)M=(Q,\Sigma,\delta,A,F)M=(Q,Σ,δ,A,F)

Where

- **Q = {A, B, C, D}**
- **Σ = {0,1}**
- **Start State = A**
- **Final States = {A, B, C}**

Final Answer

ε-Closures

- ε-Closure(q0) = {q0, q1, q2}
- ε-Closure(q1) = {q1, q2}
- ε-Closure(q2) = {q2}

DFA States

- A = {q0,q1,q2}
- B = {q1,q2}
- C = {q2}
- D = ∅

Start State

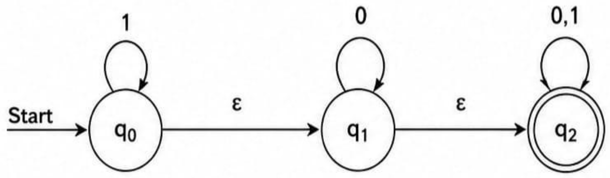
A

Final States

A, B, C

SIMATS ENGINEERING
ASSESSMENT TOOL

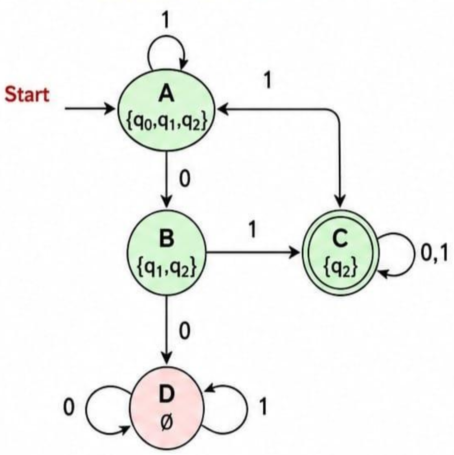
Given ε–NFA



ε–CLOSURE OF EACH STATE

State	ε–Closure
q_0	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$
q_2	$\{q_2\}$

EQUIVALENT DFA



TRANSITION TABLE OF DFA

DFA State	0	1
→* A = $\{q_0, q_1, q_2\}$	B	A
* B = $\{q_1, q_2\}$	B	C
* C = $\{q_2\}$	C	C
D = Ø (dead state)	D	D

Start State : A = $\{q_0, q_1, q_2\}$

Final States : A, B, C (All states containing q_2)

SIMATS ENGINEERING
ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand